

CSU COMPUTER-SCIENCE SENIOR PROJECT

BUCCANEER BACKROADS

Caliyah Kappel & Stefhany Mejia-Rocha – Spring, 2019



CSCI 499

Project Defense

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Statement of Purpose

The purpose of this project is to create an outlet for Charleston Southern Students to find an alternative way to interact with the city that they live in. It is apparent that many students, local and non-local are unaware of what is going on around the Charleston area. College aged students (typically 18-24) are not the targeted demographic in Charleston either, which makes it even harder to find things suitable for our age group. Buccaneer Backroads aims to alleviate the need for our demographic. There is no huge problem we wanted to solve – we just did not want students to be bored. Both members agreed that although we are both native to the Charleston area, even we are not aware of everything that goes on in the area. The Buccaneer Backroads web application not only offers food, but other activities such as shopping and group friendly things so that there is something suitable for everyone.

It will further our knowledge by learning how to appeal to certain demographics. In previous classes we did not have to have a consumer in mind when we did our projects. However, for this project everything is perceived by the students. It will teach us how to specifically focus our efforts toward a group of people. We will also learn how to effectively communicate with consumers which will be beneficial in a work environment. We will add more features to the site that sets it apart from what we've done in previous classes. By receiving constructive criticism from other students, we feel like the emotion is real and raw. Which will prepare for us for the working world better than anything else could.

Research/Background

We began by researching why exactly we needed this web application in the first place. It would serve no purpose to have a project that does not solve a want, or need. The original inspiration for the project came from the fact that we are both Commuter Students. Stefhany is from Goose Creek, and Caliyah is from North Charleston. Ironically, we found out a lot of what is going on around town from one another. Thus, the idea was born; Buccaneer Backroads was created.

To begin, we had to look at information about not only what Charleston has to offer, but the type of people in Charleston trying to cater to. Charleston is a top tourist site not only nationwide, but internationally as well. Which is great in theory, because that means there's a lot of things to do. However, these activities appeal to older age groups rather than college-aged men and women. To be a true tourist in Charleston, it costs a pretty penny too, with adults spending an average of about \$217.00 per day. The cost is another reason why we felt it was a necessity that college students had an outlet to find things to do – we are nowhere near retirement and cannot afford to do those high-end activities.

Next, came what exactly was going to be featured on the web application. We began bouncing ideas off of each other. We wanted to include variety, so a little bit of everything for everyone to enjoy. Initially, we had categories on categories which the students told us that was one thing they would cut out – they felt overwhelmed. We went back to the drawing board and came up with the one page of spotlights that featured all types of things to do rather than bogging down the page with subcategories.

We also observed how other web applications such as Yelp, Groupon, and Airbnb showcased their activities. While we wanted to have the same feel, we wanted our web

application to be a little more personal. Which means that we have tried most of the activities we list. If we don't do that, then the web application is not personal anymore and loses its flare.

Hardware/Software Required for Project

Hardware:

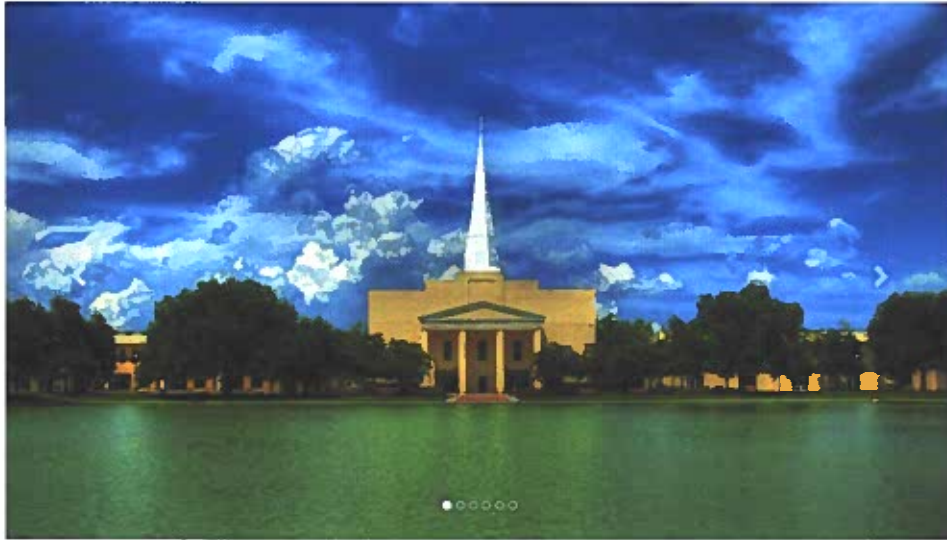
1. Stefhany's personal machine – Surface Pro
2. Caliyah's personal machine – MacBook

Software:

1. Heroku (For deployment)
 2. Amazon Web Services (AWS) Cloud 9 IDE (For Development)
 3. Bitbucket (Repository): https://bitbucket.org/CaliyahK/buccaneer_app/
 4. Michael Hartl's Ruby on Rails Tutorial
 5. Gravatar
-

Final Project Screenshots and Explanation

BUCCANEER BACKROADS



Explore Charleston, with Buccaneer Backroads

The Home page, this is what viewers see when they first open the web application. Note how in the header all they have access to is the Home and Log-in tabs, no other content.

BUCCANEER BACKROADS

Explore Charleston, with Buccaneer Backroads

You've stumbled upon Buccaneer Backroads, a website made by Charleston Southern Students, for Charleston Southern Students. Fall in love with the treasure that is Charleston, South Carolina.

Don't have an account? Sign up now!

[Sign up now!](#)

Spotlights

Here, we have placed our spotlights. The spotlight can be anything, from activities in the Charleston area, to food and fun. To truly showcase the gem that is the Lowcountry, expect to see a little bit of everything here.



Tapio: A boba tea cafe

Address: 221 Coming Street, Charleston SC



Historic Charleston City Market

Address: 185 Meeting St, Charleston, SC 29401



South Carolina Aquarium

Address: 100 Aquarium Wharf, Charleston, SC 29401



Ceviche and Bananas

Address: 51 George St, Charleston, SC 29401



Rodney Scott BBQ

This is a continuation for the home page, it informs viewers about the web application, and provides a button for users who do not have an account to sign up. Underneath, is a masonry grid layout of things to do, just in case a viewer wants a peek at what the web application is designed for. They still cannot view other content.

Contact Us!

If you have any questions, comments, or concerns, please do not be afraid to contact us! We will get to your email as soon as possible, and respond within a reasonable amount of time.



Stefany Mesa

smesaroche@csustudent.net

B.S. Cybersecurity



The contact page provides the contact information for the creators. (Located in the footer)

About

Hello! Our names are Stefany Mesa-Rocha and Caliyah Kappel. Welcome to Buccaneer Backroads! We're both seniors at Charleston Southern University, and we also are both majoring in cybersecurity.

Our Vision

We are both Charleston natives, so this website is very important to us. We realized going into our senior year of college, that a lot of students (including us) don't know what is going on around town most of the time. So, we decided to create a website that is for people around our age (18-24) that shows them things to do around town. Charleston tourist sites do not target our demographic, and also does not show the many cool things to do in the area. By having a website made by students for students, we're hoping to show in and out-of-state students the many wonders of Charleston, South Carolina!

This website is for our HCI (Human Computer Interaction) class for the Fall of 2018. We're hoping that this website displays everything that we've learned over the course of the semester. There were many factors that were taken into consideration while building this for our project. We began with an app, but decided to stick with what we know and code in Ruby! We hope after graduating that this continues to flourish and hopefully be adopted by the school.

Not only did we create this website for our HCI class, we decided to further adapt it for our senior project! We really fell in love with the idea of having something that the students could interact with, so we added some new things for the website.

For any viewer curious as to what the web application is about, they can visit the about page which talks about the reasoning behind the web application. (Located in the footer)

Log in

Email

Password

☐ Remember me on this computer

Log in

[New user? Sign up now!](#)

Sign up

Name

Email

Password

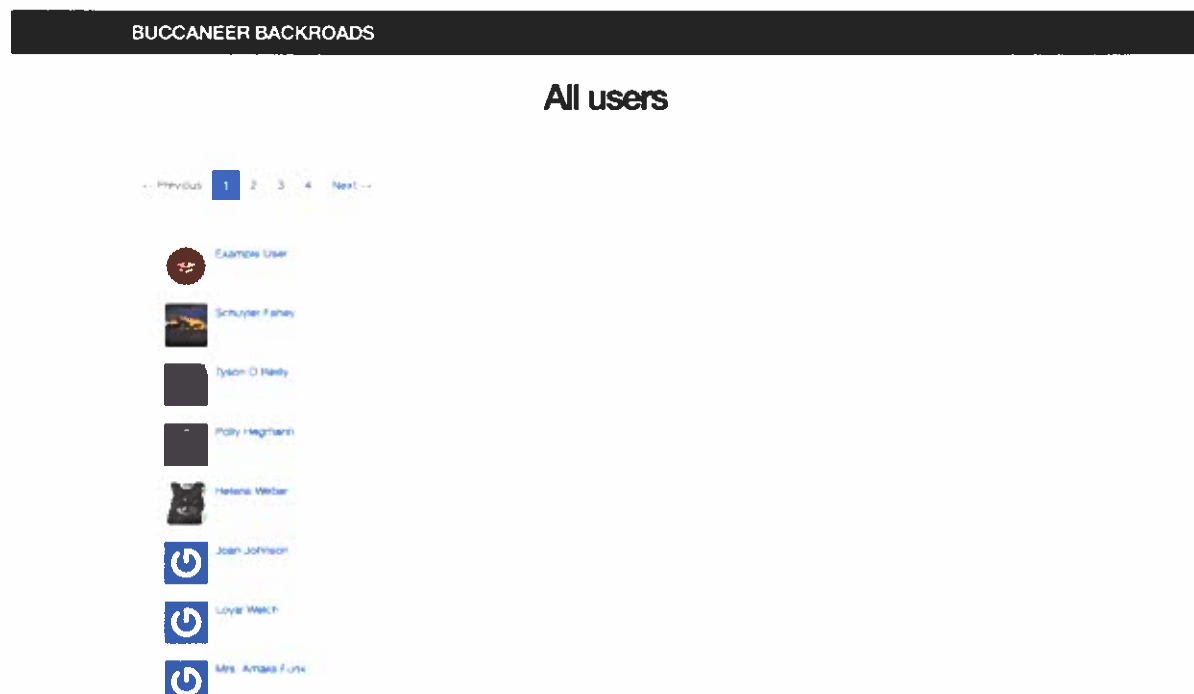
Password confirmation

Create my account

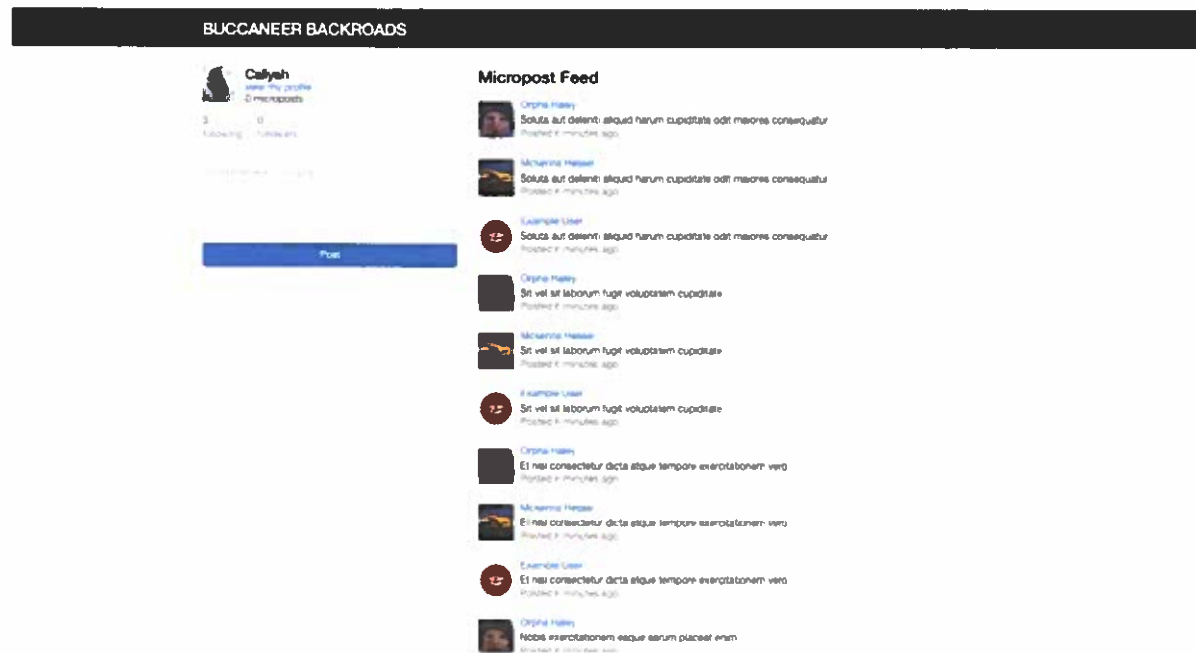
The log-in and sign-up pages, they have fields for the user to enter a name, email and password. Certain constraints have been put in place to prevent multiple accounts under one email, prevention of leaving fields blank, and password length to prevent buffer overflow attacks.



Once the user successfully signs up, they will be welcomed to the web application. Observe the header now, they can look at other users, explore the other options of things to do, and manage their account.



This page displays all users that are on the web application, they are fake users, and we used the faker gem to populate this page.



This is how the user's feed appears, it displays the micro posts of users that they follow, and allow for the user to make their own micro post. Also, it shows their statistics of how many followers they have, and how many users they follow.



This is the feed of a user; the top is where the follow button is located. Once clicked, the button changes to unfollow just as it would a regular social media platform.

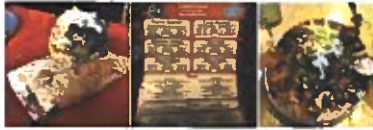
Late Night Eats

Just spent the last 5 hours cramming for your midterm exam? Just passed it's almost midnight, but you want to celebrate, your late night cravings? Or just bring a few beers to eat at with some friends on a late weekend night? We got it, you're tired of having Wendy's and McDonald's every night since they seem to be the only restaurants open late. Here not, we've got you back! Check out these late night eats, I've put out by our very own CSU students! Cheap, AND handy for your college budget, a match made in heaven.

Juanita Greenberg's Nacho Royale



Cravings? Check. Tacos? Check. Nachos? Check. Chees? Check. Juanita Greenberg's Nacho Royale is your one stop shop for delicious Mexican eat' food. From saucy quesadillas to sky high' plast nachos, the menu has a little something for everyone. After a crazy week, there's nothing like a delicious burrito to fill you up, and cheer you up.



Some popular menu items include:

1. Taco Plate: Any 2 tacos, rice, and cheese beans
2. Quesadilla: Flour tortilla, pizza, and Monterey Jack cheese
3. Carnitas: Chops with black beans, pico de gallo, lettuce, and queso

View the entire menu: [here](#)

The Late Night Eats page is exactly as it sounds, it has its own spotlight and if you scroll further it lists three other places that you can eat late night food. There is a description for the spotlight, and the setup is the same for the Thriller Page.

Thrillers

Are you a true thriller? Adventure lover? Or just have some extra time for something fun to do? First of going to the movies every Friday night? We have some fun, fast, heart pumping, hair raising fun. Race against the clock, facing a series of brain teasers, puzzles, and your friends.

Escape Room Games



88th Escape Games

88th Escape Games has 4 different escape rooms, each giving a time limit of 60 minutes. There is a room for everyone, depending on the participant's common level. Not every one's a genius? That's fine, try out the "Terrorist" room. Sometimes you're the "Terrorist" room, where you are being held hostage in a haunted subway car, to "Silhouette" where you attempt to prevent a nuclear meltdown from occurring. Run away from aliens in "Escape Area 51" and confront an outbreak of a brain eating virus that can turn people into zombies in "Undead Outbreak". These rooms are sure to test your brain!

Finished all the rooms but you're still wanting more?

You can also visit

[Escape in 60](#) and

[Break Out Charleston](#)

to see Thrillers.

Wild Blue Pigeons

The blue pigeons come if for
amazing photo taking to an
amazing photo taking to an

Urban Adventure Quest

Explore Charleston in this
adventure hunt that spans
the city

Hydroly Water Sports

Charleston can get pretty hot in
the summer months. Hydroly
water sports are a great way to
cool off

This is the thriller page, and it has a spotlight of escape room games.

BUCCANEER BACKROADS

Beaches

Not only is Charleston known as the Holy City, but this gem also has its fair share of beaches! The lowcountry beaches provide breathtaking views that will stop anyone dead in their tracks to admire them. Folly Beach and Isle of Palms promise to provide beautiful scenery and a fun filled day for any beach goer.



Lastly is the beaches page. For the beaches page, there is a general description of the beaches, and provides an image carousel for the users to browse and look.

Test Plan

User Evaluation Test Plan

Buccaneer Backroads

March 1st, 2018

Version 2.0

Stefhany Mejia-Rocha

Caliyah D. Kappel

1. Purpose and Objectives

Purpose — The purpose of the user evaluation study is to evaluate the time required for users to navigate through different pages of the web application. The operators will be asked to find a total of three pages. After that, they will be encouraged to give feedback on what they like and dislike.

Objectives — Objectives for the activity are:

1. To identify and locate three activities.
2. To navigate through each page.
3. To identify an activity, they would enjoy or foresee themselves participating in.

Hypotheses – Hypotheses for the study are:

1. If the user is able to successfully complete the tasks in a reasonable amount of time, then the web application will be effective for its purpose.
2. If the user is able to navigate through the pages in a reasonable amount of time, then the user interface of the web application will be widely accepted by other users.
3. If the user is able to find an activity they are interested in and would like to participate in, then the web application serves its purpose in benefitting college students.

2. Test Procedure

Summary of procedure — The plan for conducting the user evaluation study and collecting data is to introduce the users to the Buccaneer Backroads web application, provide instructions to complete a given task, collect data for the time required to complete the tasks, collect demographic data, compile findings, and determine conclusions.

Place and time — Each participant will use the Buccaneer Backroads web application individually. The session will be held in Java City coffee shop on Charleston Southern University's campus. The usability test is planned for twenty minutes for each participant.

3. Participants

Test participants — Twenty Charleston Southern University students will be the participants in the study. No one has an advantage over another user because the users are chosen at random.

They should have the following skills, experience, and characteristics:

- Has to be enrolled at Charleston Southern University (at least part-time).
- Various levels of experience.
- No specific major is required to participate.
- Has to be between the ages of 18 to 24.
- Normal dexterity with at least one hand, either right or left, adequate for using a mouse cursor positioning device.
- No specific gender requirement
- Willingness and enthusiasm to freely give opinions about good and bad features of the web application being used and tasks being performed.

Compensation — There will be no compensation for participants in this study.

4. Site

Requirements for facility — The site for the time on task study will be the **Java City** coffee shop on Charleston Southern University's campus. This location will have the following capabilities:

- We will bring our own equipment (laptop, charger, pens, paper).
- The only thing needed from our site (from a technical aspect) will be access to their Wi-Fi.

5. Experimental Design

Independent variables — We will categorize the participants according to the following independent variables:

- Major/Minor.
- Gender—male, female.
- Age range.

Dependent variables — Quantitative performance measures for the validation testing will include some or all of the following:

- Time required to complete locating specific assignments on the web application.
- Time required to navigate through each page.
- Time required to find an activity that peaks the interest of the user.

6. Resources

Materials — Materials needed for the test include:

- Laptop
- Charger
- Pen
- Paper
- Wi-Fi access

7. Data Collection

Sources of data — The sources of data for time on task data collection are:

- Excel document that serves as a database for all participants in the study (Name, ID)
- Background demographic (age, major) questionnaires.

Data topics - The data collection topics will include:

- Time for completing an assigned task.
 - Time for navigating through each page.
 - Time needed for a user to find an activity that they are interested in.
-

8. Data Analysis

Purpose — The purpose of data analysis is to determine the time required to complete the tasks given to each user. The sample size is not large enough to apply a set conclusion with our findings, but some statistics will provide what we need to improve on in our final product.

Activities — Data analysis activities are:

- Review time it takes for users to complete a task.
- Calculate average amount of time for users to navigate through each page.
- Compile statistics of the data, such as means, and graph the results.

9. Deliverables

The following products of the time on task study will be delivered:

- Timing data—raw data, compilation, statistics, and graphs.
- Report describing results and conclusions.

10. Schedule

Schedule for data collection — The schedule for conducting the data collection is shown below.

Orientation	5 minutes
Exercises	10 minutes
Questionnaire/debriefing.....	5 minutes
TOTAL.....	20 minutes

11. Trial Scenarios

Timing Task: Successfully locate the page that has been assigned, successfully navigate through each page, and find an activity that the user wants to interact with.

The operator will be asked to locate three pages. The timing will begin when the operator begins to locate each page.

General Orientation

"We are conducting a study on a web application called Buccaneer Backroads. Buccaneer Backroads is a web application designed to provide activities and eateries that college students would find interesting. We invited you to participate in this study because you represent the type of user who might benefit from a web application like this."

"Today, you will use and navigate through the web application. The purpose of the session is not to evaluate your performance but to evaluate the web application."

"You will use our personal laptop. You should not need to use the keyboard."

"I will ask you to navigate and find three specific activities. If at any time during the session you feel uncomfortable or that you are unable to continue, you may discontinue your participation."

"Please feel free to say out loud what you are thinking. You should offer as many comments as you can about what you're doing, what you see on the screen, and your opinion about good and bad features. If you don't know what action to take, we can give you some advice, but we won't offer any help unless you really need it."

"I'll spend a few minutes now to explain the different aspects of Buccaneer Backgrounds. Do you have any questions before getting started?"

Background Information

Name (optional): _____

Gender: ☐ Female ☐ Male

Age: _____

Major/Minor: _____

How long have you been at Charleston Southern University?: _____

Freshman/Sophomore/Junior/Senior: _____

Buccaneer Backroads

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Abstract—This paper discusses our Senior Project, **Buccaneer Backroads**. It is made for Charleston Southern University (CSU) students to find things to do outside of campus. We conducted user evaluations by letting students chosen at random go through the web application and then answer a survey on what they like and/or disliked about the web application. Once the web application was finished, we conducted the user evaluation again to see how our results changed from the prototype. Students thoroughly enjoyed the product and are more apt to use an application that cater to their interest(s). Students find it easy/simple to interact with the web application, it is straightforward and does not overwhelm them.

Keywords— *Charleston Southern Students, student engagement, Charleston, activities.*

Introduction

Buccaneer Backroads helps alleviate stress and boredom for college students to explore new places around the Charleston area. There are not many widely available web applications or applications that show activities to do around the city for college students. This web application is designed to encourage Charleston Southern University students to interact with their community and try out new things. Not only do students benefit from engaging in this application, but so do other young adults that are looking for something new to try and do. The surveys, questionnaires, and studies conducted for this web application are important because it demonstrates the effectiveness of the web application.

The reasoning behind this web application came from us realizing that we never take the time to be a tourist where we live. Seeing this, we deduced that if we do not explore Charleston on a regular basis, then we know that the students at Charleston Southern (more of the out-of-state or non-local) students definitely would not know of the things to do around the area. Not only that, but we have seen that the main demographic targeted was the older demographic. Most college kids are not interested in the same things as the older generation. This way, they (college students) can find something that is fun to do and more in agreeance with their age range (about 18-24).

System Description

Interface Design

The interface design was somewhat simple in planning. We did not want the pages to overwhelm the user, while also not leaving pages so bare that there was void space. The home page has a centered image at the top of the page. Following the picture, there is a small paragraph that discusses the reasoning behind the web application. There is also a flexbox underneath the description that serves as a blog. The web application does not vary from page to page with regards to the color scheme. The color scheme is composed of warm and cool blues, followed by either white or gray font depending on the backdrop.

The other pages that had content with actual activities were formatted slightly differently. Under the first header there is a slight description to match the page's content. Simply put, it ensures that the type of food and/or activity corresponds with the

page's content. Following the description is a "spotlight" of some sort that is followed by the highlights of the activity/eatery. Each picture on a tab is a hyperlink, so if the user would like to find out more about the activities then they can go directly to the web application that is featured.

Apparatus and methods

Participants

Participant requirements:
They should have the following skills, experience, and characteristics:

- Has to be enrolled at Charleston Southern University (at least part-time).
- Various levels of experience.
- No specific major is required to participate.
- Has to be between the ages of 18 to 24.
- Normal dexterity with at least one hand, either right or left, adequate for using a mouse cursor positioning device.
- No specific gender requirement
- Willingness and enthusiasm to freely give opinions about good and bad features of the web application being used and tasks being performed.

Numb er	Who the Participant is	Importa nt Screenin g Criteria	Demographics (Gender/Age/Class)
------------	------------------------------	--	--

#1	CSU Undergradu ate	CSU Student	Male/22/Senior
#2	CSU Undergradu ate	CSU Student	Male/20/Sophomor e
#3	CSU Undergradu ate	CSU Student	Female/20/Sophom ore
#4	CSU Undergradu ate	CSU Student	Male/19/Freshman
#5	CSU Undergradu ate	CSU Student	Male/20/Freshman
#6	CSU Undergradu ate	CSU Student	Female/22/Senior
#7	CSU Undergradu ate	CSU Student	Female/19/Sophom ore
#8	CSU Undergradu ate	CSU Student	Female/18/Freshm an
#9	CSU Undergradu ate	CSU Student	Male/18/Freshman
#10	CSU Undergradu ate	CSU Student	Male/24/Senior
#11	CSU Undergradu ate	CSU Student	Male/18/Freshman
#12	CSU Undergradu ate	CSU Student	Male/19/Sophomor e
#13	CSU Undergradu ate	CSU Student	Female/18/Freshm an
#14	CSU Undergradu ate	CSU Student	Male/21/Junior
#15	CSU Undergradu ate	CSU Student	Female/19/Sophom ore

#16	CSU Undergraduate	CSU Student	Female/20/Junior
#17	CSU Undergraduate	CSU Student	Female/19/Sophomore
#18	CSU Undergraduate	CSU Student	Male/18/Freshman
#19	CSU Undergraduate	CSU Student	Female/21/Senior
#20	CSU Undergraduate	CSU Student	Male/22/Senior

pen and paper in order to collect data from the study. A total of 20 Charleston University students participated in this study. Each person was given a specific task while navigating the web application. Every participant was given the same task and were chosen at random. Participants did not need any prior knowledge or training prior to the study. The tasks were given as follows:

1. User will be asked to locate three specific pages.
2. User will be asked to navigate through each page.
3. User will be asked to find an activity that he or she has an interest in.

Experimental Design

We have categorized the participants according to the following independent variables:

- Gender – male, female.
- Age range.

Quantitative performance measures for the validation testing will include all of the following dependent variables:

- Time required to complete locating specific assignment of the web application.
- Time required to navigate through each page.
- Time required to find activity that peaks the interest of the user.

Objective metrics include time required for the tasks, and the subjective metrics include the attitude towards the web application, feedback on the web application's performance, and its quality.

Data was collected while the participants were performing their tasks by using a timer on a cellphone.

Apparatus

The study was conducted online using Charleston Southern University's Internet connection at Java City. The Buccaneer Backroads web application was pulled up onto our personal laptops. We also used

Results

The following is the results from our closed question interview conducted in Java City on the Charleston Southern University Campus. The participants were 20 students that were chosen at random to prevent bias from occurring. This was so we were able to gauge our audience. Here are the questions that we asked the participants:

1. Do you wish there were more apps that offered activities around Charleston for our age group?
2. If this app existed, would you recommend it to friends?
3. Does Charleston Southern provide sufficient activities to do outside of school?
4. How important are price ranges in food menu items when you go out to eat? (Scale of 1-5, 1 being extremely unimportant)
5. On a scale of 1 to 5 (with 1 being extremely unimportant; 5 being extremely important), how important is the location's distance from the campus?
6. Do you look at ratings before you go somewhere?
7. Are you open to exploring new experiences out of your comfort zone?

8. Do you enjoy religious activities?

9. Do you have a hard time finding things to do on the weekend?

10. Are you a native Charlestonian?

From our interview questions, we received the following answers. We also included why we asked the questions. The answers are in order corresponding to the question number.

For question 1:

18 out of 20 or 90% said yes, they wish that there were more activities offered in the area for our age group.

- We wanted to incorporate this question to see if our application will be beneficial to college students at CSU.

For question 2:

12 out of 20 or 60% said yes, they would recommend the app to friends.

- We asked this question to see if the application would maintain its purpose and success.

For question 3:

19 out of 20 or 95% said no, Charleston Southern does not provide sufficient activities to do outside of school.

- We asked this question to ensure the creation of the application and its usefulness.

For question 4:

16 out of 20 or 80% voted 4 to 5 on importance of food menu items when they go out to eat.

- We asked this to brainstorm a selection of restaurants, regarding the majority of the student's votes.

For question 5:

14 out 20 or 70% voted 4 to 5

on importance of location's distance from the campus?

- We asked this to brainstorm restaurants, activities, and events, regarding the majority of the student's votes.

For question 6:

10 out of 20 or 50% voted yes, they look at ratings before they go somewhere.

- We asked this because we wanted to know the importance of ratings and if we should incorporate our own personal rating system for the activities.

For question 7:

- 16 out of 20 or 80% voted yes, they are willing to explore new experiences out of their comfort zone.

- We asked this to broaden the user's horizons if they were willing to try new experiences.

For question 8:

12 out of 20 or 60% voted yes, they enjoy religious activities.

- We asked this to see if we should incorporate religious activities in the application.

For question 9:

15 out of 20 or 75% voted yes, they have a hard time trying to find things to do on the weekend.

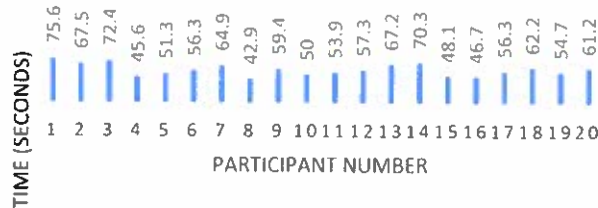
- We asked this question to get an idea of popular weekend activities we can include for the application.

For question 10:

12 out of 20 or 60% voted yes, they are native Charlestonians.

- We asked this question because we were curious to see if Charlestonians would use an application that is based on activities where they live.

TIME TAKEN FOR A USER TO COMPLETE A GIVEN TASK

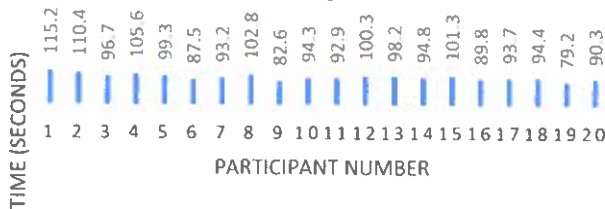


Average: 58.19

Median: 56.8

Standard deviation: 9.38

TIME TAKEN FOR A USER TO NAVIGATE SPECIFIC PAGE

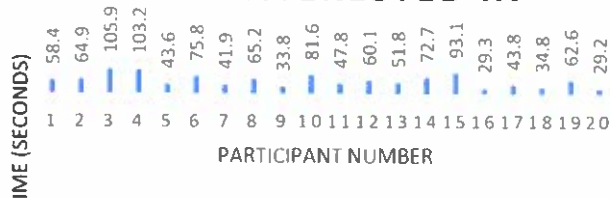


Average: 96.125

Median: 94.6

Standard deviation: 8.62

TIME TAKEN FOR USER TO FIND SOMETHING THEY ARE INTERESTED IN



Average: 59.975

Median: 59.25

Standard deviation: 23.28

Discussion/Conclusion

The results from our study did match our hypotheses very closely. We stated that the three following hypotheses:

1. If the user is able to successfully complete the tasks in a reasonable amount of time, then the web application is effective for its purpose.
2. If the user is able to navigate through the web application in a reasonable amount of time, then the user interface of the web application will be widely accepted by other users.
3. If the user is able to find an activity that they are interested in and would like to participate in, then the web application serves its purpose in benefitting college students.

Many students after walking through the web application, said that with the UI and the feel of it would continue using it and would more likely than not say something to their friends about it. Based on the feedback, many students did want to adopt the blog layout for themselves. This way they are able to interact with other students, and post their likes and/or dislikes if they have eaten at that restaurant or participated in a certain activity.

One of the main implications that came up that we saw was if students didn't like a certain aspect of the web application then they became more likely to not want to interact with the web application at all. Because of this, we really focused in on making the web application appear and function at its highest capacity. One thing that we could also add is error prevention. We don't expect a majority of the users to understand what they're looking at if an error code comes up while the user is engaged with the system. We can use language that is easy to understand so that the user can comprehend what they are looking at.

Flexibility and efficiency of use can be useful to apply to our web application. By incorporating this heuristic, we can implement shortcuts to different tabs so the user can quickly find what he/she is interested in. Tabs are an easy way to make this happen, because the user can use these shortcuts to directly visit the part of the site. On top of that, we have incorporated links into the images on the web

application so the user can simply click the photo to head directly to the desired web application.

Challenges Faced or Overcome

Learning to take criticism in stride and change what we had at the drop of a hat was one of the biggest challenges. We wanted to provide students with a multitude of things to do, so we had many options to choose from. What we did not foresee was students saying that they felt overwhelmed by the amount of pages that were available. While we did not have to restart the idea of the project from square one, we had to decide how we wanted to format the web application to agree with the consumer (in our case, students). In classes, for projects we are given a direct outline of how the professor wants the appearance, code, functionality, and format to appear. This was different; we were allowed to roam freely and face students who do not have a background in computer science. At some points it proved to be challenging – their recommendations were outside of our scope, and we had to tell them such without sounding rude.

Not being able to implement the mailer into our project for account verification was something that we really wanted to do but were not able to. Essentially, the user was supposed to be able to sign up and then would be sent an account verification link in order to activate their account. What happened was that the web page would say that the link was sent out to the user, but the user never received the e-mail. It made it to where we were unable to log in and access the content because we could not verify the account. While it was something that we were able to code in, it did not execute the way that we wanted which was frustrating.

We wish that we were able to make the site more interactive for the users. This includes things like rating systems, and messaging between users. Taking this project further, these are

definitely aspects that should be implemented into Buccaneer Backroads. We think that students would really appreciate something that is interactive.

Another challenge was putting everything in the web application that both members wanted. We found ourselves bumping heads a few times about tab placement, color scheme, and other things that would make each member happy. This seems like something small now, because we were able to muscle through the disagreements and this helped in the long run. Because of our field, many times we will be expected to work in teams and communicate any questions, comments, or concerns clearly and quickly. Not all experiences will be easy, and this project helped better prepare us for when we inevitably face this situation in the workplace.

Another feature we wanted to add was the password reset feature – this would allow for the user to receive an email with instructions on how to reset their password. We were not able to implement this feature because of the mailer, as talked about earlier. This, we would have to say is one of our security flaws in the web application. Users don't receive an activation link, which means someone can use anybody's email, and users cannot reset their passwords which means there is no current way for them to get back into the system with the email they used.

Time was not our friend for this project – we had the idea in our heads but we needed to make our idea a reality, in about a semester. We doubled up on the project and defense, because we wanted to graduate on time, so we had no time off. Biweekly meetings with Dr. Hayes kept us on track, but sometimes we found ourselves questioning if we had added enough content for each meeting. Adding a lot of static pages would bite us later with showing the project to students.

Future Enhancements

The significance of our findings were clear – students wanted a web application that they could interact with. If the web application was to their liking, not only could they see themselves using it on a regular basis, but they would also recommend it to their friends. This would create more interest in the web application and eventually (hopefully) would lead to multiple people using the web application across the campus.

In the future, we hope that with further implementations such as a mobile version, Charleston Southern University adopts to use. Only because we know that this is something that students want, and a web application such as Buccaneer Backroads is a web application that can put Charleston Southern ahead of its competition. This web application will be a first of its kind in the Charleston area and for most colleges for that matter that is unique to the students there. You can compare this to the likes of yelp, but it has its own flare and own personality.

If limitations were identified, we would not completely scrap the web application that we have created. Rather, we would fix any issues, and then see if the problem causes clashes in other areas of the web application. We hope to not face any of these obstacles, since we do plan on further developing this project in the near future.
